

legged_mani: B2-Z1 Push-Box Controller

Mathematical Formulation and Implementation Analysis

Notes based on analysis of the `legged_mani` codebase

July 29, 2026

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1 Scope and Executive Summary

This document analyzes the `push_box` task implemented by `control/controllers/mppi_push_box.py`. The controller moves a free box toward a fixed world-frame goal with the end effector (EE) of a four-degree- of-freedom Unitree Z1 arm mounted on a Unitree B2 quadruped. The analysis is based primarily on:

- push-box task logic and rollout cost: `control/controllers/mppi_push_box.py`,
- shared sampling and whole-body support: `control/controllers/base_controller.py` and `control/controllers/whole_body_arm_controller.py`,
- kinematic base-pose planning: `control/planning/base_pose_cem.py`,
- task and controller parameters: `utils/tasks.py` and `configs/mppi_push_box.yml`,
- rollout and simulation models: `models/b2_z1_base_push_box.xml` and `models/scene_push_box.xml`.

The controller combines two optimizers with different roles:

1. A background cross-entropy method (CEM) searches a three-dimensional kinematic base-pose reference $[\Delta x_b, \Delta y_b, z_b]^\top$ that makes the moving box-contact point reachable.
2. Whole-body MPPI samples 16-dimensional desired joint-position trajectories and evaluates their full robot–box contact dynamics in parallel MuJoCo rollouts.

The CEM result is only a reference. MPPI remains the sole dynamic trajectory optimizer. Its cost combines robot state tracking, virtual PD effort, box position and tilt, moving EE-to-box contact tracking, and arm–torso clearance. The contact point changes from a pre-contact point outside the box to a small virtual penetration inside the pushing face after engagement.

2 Robot, Box, State, and Input

2.1 Degrees of Freedom

The push-box model contains the 23 robot position coordinates and 22 robot velocity coordinates of the B2–Z1 system, plus a seven-coordinate free-joint pose and six velocities for the box:

$$n_q = 30, \quad n_v = 28, \quad n_u = 16. \quad (1)$$

The actuated coordinates are the twelve B2 leg joints and the first four Z1 arm joints:

$$\mathbf{q}_j = [q_{FR,h}, q_{FR,t}, q_{FR,c}, q_{FL,h}, q_{FL,t}, q_{FL,c}, q_{RR,h}, q_{RR,t}, q_{RR,c}, q_{RL,h}, q_{RL,t}, q_{RL,c}, q_{a1}, q_{a2}, q_{a3}, q_{a4}]^\top.$$

Here h, t, and c denote hip, thigh, and calf.

2.2 Observation State

The controller receives the complete MuJoCo state $[\mathbf{q}_{pos}^\top, \mathbf{q}_{vel}^\top]^\top$ and separates it into a canonical 45-dimensional robot state and a 13-dimensional box state:

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_r \\ \mathbf{x}_o \end{bmatrix} \in \mathbb{R}^{58}, \quad (2)$$

$$\mathbf{x}_r = \begin{bmatrix} \mathbf{p}_b \\ \mathbf{q}_b \\ \mathbf{q}_j \\ \mathbf{v}_b \\ \boldsymbol{\omega}_b \\ \dot{\mathbf{q}}_j \end{bmatrix} \in \mathbb{R}^{45}, \quad \mathbf{x}_o = \begin{bmatrix} \mathbf{p}_o \\ \mathbf{q}_o \\ \mathbf{v}_o \\ \boldsymbol{\omega}_o \end{bmatrix} \in \mathbb{R}^{13}. \quad (3)$$

The base and box quaternions use MuJoCo’s scalar-first convention $[q_w, q_x, q_y, q_z]^\top$. The robot state is formed by excluding the box free-joint indices from the full arrays; the box state is extracted using the `box_joint` position and velocity addresses. This is safer than assuming that the two subsystems always occupy fixed contiguous blocks.

The MuJoCo FULLPHYSICS rollout buffer additionally stores time:

$$[t, \mathbf{q}_{\text{pos}}^\top, \mathbf{q}_{\text{vel}}^\top]^\top \in \mathbb{R}^{59}. \quad (4)$$

The leading time element is removed before cost evaluation, leaving the 58-dimensional observation above.

2.3 Control Input and Actuator Model

The optimized input is a vector of desired joint positions, not joint torques or contact forces:

$$\boxed{\mathbf{u} = \mathbf{q}_{j,\text{des}} \in \mathbb{R}^{16}}, \quad u_i^{\text{clip}} = \text{sat}_{[u_i^-, u_i^+]}(u_i). \quad (5)$$

Each affine MuJoCo position actuator then computes

$$\tau_i^{\text{unsat}} = k_{p,i}(u_i^{\text{clip}} - q_{j,i}) - k_{d,i}\dot{q}_{j,i}, \quad (6)$$

$$\tau_i^{\text{MJ}} = \text{sat}_{[\tau_i^-, \tau_i^+]}(\tau_i^{\text{unsat}}). \quad (7)$$

The hip actuators use $(k_p, k_d) = (100, 5)$, the thigh and calf actuators use $(300, 8)$, and the four arm actuators use $(300, 5)$. Desired positions are clipped by the sampler, while MuJoCo applies the separate actuator-force limits.

3 Prediction Dynamics and Contact

3.1 Discrete MuJoCo Dynamics

One prediction step is the full contact-rich transition

$$\boxed{\mathbf{x}_{k+1} = F_{\text{MJ}}(\mathbf{x}_k, \mathbf{u}_k; \boldsymbol{\theta}_{\text{model}})}. \quad (8)$$

The transition includes floating-base and articulated rigid-body dynamics, position-PD actuation, force saturation, ground contact, and physical EE–box contact. The rollout model disables only physical contact between the arm subtree and the torso collision geometry; their clearance is instead handled by the analytic soft cost and hard rejection described later. Arm contact with the box and ground remains enabled.

The configured integration interval and horizon are

$$\Delta t = 0.01 \text{ s}, \quad H = 40, \quad T_H = H\Delta t = 0.4 \text{ s}. \quad (9)$$

The rollouts use a pyramidal friction cone and the enabled contact override

$$\mathbf{o}_{\text{solref}} = [0.02, 1.0]. \quad (10)$$

The simulator copies the rollout model’s cone, override flag, `o_solref`, `o_solimp`, friction override, and margin so that predicted and executed contact options match.

3.2 Implicit Contact Scheduling

No contact force appears in $\mathbf{u}$, and there is no prescribed stance, swing, or EE–box contact mode in the dynamic optimization. Leg contact is determined by the height-conditioned gait prior together with MuJoCo collision. Box pushing emerges from physical arm–box contact when sampled joint-position commands bring the EE into the box.

The Boolean `contact_engaged` flag does not measure a MuJoCo contact force. It switches when the measured EE position is sufficiently close to a geometric pre-contact target. It changes the reference used by IK and the cost, but physical contact is still resolved exclusively by MuJoCo.

4 Finite-Horizon Stochastic Optimal Control Problem

For a fixed reference supplied by the task logic and CEM planner, the intended MPPI problem can be written as

$$\begin{aligned} \min_{\mathbf{U}} \quad & J(\mathbf{U}; \mathbf{x}_0) = \sum_{k=0}^{H-1} \ell_k(\mathbf{x}_{k+1}, \mathbf{u}_k) + \Phi_{\text{EE}}(\mathbf{x}_H), \\ \text{s.t.} \quad & \mathbf{x}_{k+1} = F_{\text{MJ}}(\mathbf{x}_k, \mathbf{u}_k), \\ & \mathbf{u}_{\min} \leq \mathbf{u}_k \leq \mathbf{u}_{\max}, \\ & d_{k,s} \geq d_{\text{hard}}, \quad s = 1, 2, 3, \\ & \mathbf{x}_0 = \hat{\mathbf{x}}(t), \end{aligned} \tag{11}$$

where $\mathbf{U} = [\mathbf{u}_0, \dots, \mathbf{u}_{H-1}]$ and $d_{k,s}$ is the arm-capsule-to-torso clearance. The box goal is a soft objective, not a hard terminal constraint.

The controller samples $N = 30$ candidate sequences around a structured gait/IK prior. In parallel, the CEM planner periodically solves a separate static kinematic problem for the base reference. CEM does not propagate dynamics, score box motion over the MPPI horizon, or replace the MPPI optimization in Eq. (11).

5 Push Geometry, Base-Pose Planning, and Task Progression

5.1 Box Goal and Push Direction

The single configured box-position target is

$$\mathbf{p}_o^* = [3.0, 2.0, 0.19]^\top \text{ m}. \tag{12}$$

For a predicted box position $\mathbf{p}_{o,k}$, the horizontal push direction is

$$\mathbf{d}_k^{xy} = \frac{\mathbf{p}_o^{*,xy} - \mathbf{p}_{o,k}^{xy}}{\|\mathbf{p}_o^{*,xy} - \mathbf{p}_{o,k}^{xy}\|}, \quad \mathbf{d}_k = \begin{bmatrix} \mathbf{d}_k^{xy} \\ 0 \end{bmatrix}. \tag{13}$$

When the horizontal distance is numerically zero, the most recent valid push direction is reused.

5.2 Moving Contact Target

Let $\mathbf{R}_{o,k}$ be the world-from-box rotation and $\mathbf{h}_o = [0.19, 0.19, 0.19]^\top \text{ m}$ the box half-size. The distance from the box center to its surface along $\mathbf{d}_k$ is

$$\rho_k = \min_{i: |(\mathbf{R}_{o,k}^\top \mathbf{d}_k)_i| > 0} \frac{h_{o,i}}{|(\mathbf{R}_{o,k}^\top \mathbf{d}_k)_i|}. \tag{14}$$

The moving EE target lies on the face opposite the desired motion:

$$\tilde{\mathbf{p}}_{e,k}^* = \mathbf{p}_{o,k} - \rho_k \mathbf{d}_k + \delta \mathbf{d}_k, \tag{15}$$

after which its vertical component is explicitly overwritten as

$$p_{e,k}^{*,z} = p_{o,k}^z - 0.05 \text{ m}. \quad (16)$$

The offset δ defines two reference modes:

$$\delta = \begin{cases} -0.05 \text{ m}, & \text{before engagement,} \\ +0.015 \text{ m}, & \text{after engagement.} \end{cases} \quad (17)$$

Thus the first target stays outside the rear face, whereas the second is a small virtual penetration into the face that encourages sustained physical contact. Every rollout recomputes this target from its own predicted box pose; the cost does not track only the currently observed target.

Engagement occurs when the current EE is within 0.04 m of the pre-contact target. The switch is one-way until success and triggers a CEM refresh for the new penetration target.

5.3 Success Condition

Only horizontal box error determines task completion:

$$\|p_o^{xy} - p_o^{*,xy}\| \leq 0.20 \text{ m}. \quad (18)$$

The inequality must hold for 20 consecutive calls to `goal_reached`; the counter resets to zero whenever the box leaves the tolerance. On success, pending CEM work is invalidated, contact engagement is cleared, the `stance_hold` gait is selected, and the controller holds the current base position.

5.4 Kinematic CEM Base-Pose Planner

5.4.1 Decision Variable and Sampling

The asynchronous planner searches

$$\mathbf{z} = [\Delta x_b, \Delta y_b, z_b]^\top \quad (19)$$

using a private MuJoCo model, private IK data, and a dedicated random generator. Its feasible search region is

$$\|[\Delta x_b, \Delta y_b]^\top\| \leq 0.40 \text{ m}, \quad 0.35 \leq z_b \leq 0.543542 \text{ m}. \quad (20)$$

CEM draws 30 samples per iteration for three iterations, retains the best 20% as elites, and exponentially smooths the new distribution with factor 0.15. The first iteration also includes deterministic seeds at three forward offsets and three height levels. A complete plan normally evaluates $30 \times 3 = 90$ kinematic candidates and launches no additional MPPI rollout.

5.4.2 CEM Objective

For a candidate base pose, position-only IK produces an arm posture and EE residual r . Define

$$\eta_r = \frac{r}{0.03}, \quad v_r = \max(\eta_r - 1, 0), \quad (21)$$

$$\eta_{xy} = \frac{\|[\Delta x_b, \Delta y_b]^\top\|}{0.40}, \quad \eta_h = \frac{0.543542 - z_b}{0.543542 - 0.35}. \quad (22)$$

Let ℓ_{clr} denote the normalized arm-torso clearance deficit and ℓ_{lim} the normalized arm joint-limit margin deficit. The implemented CEM score is

$$J_{\text{CEM}} = \eta_r^2 + 100v_r^2 + 1000\mathbb{I}_{\text{infeasible}} + 6\eta_{xy}^2 + 3\eta_h^2 + 2\ell_{\text{clr}} + \ell_{\text{lim}} + 10^6\mathbb{I}_{\text{arm collision}}. \quad (23)$$

A CEM candidate is labeled feasible only when $r \leq 0.03$ m and the arm–torso hard-clearance test passes.

Before a CEM pose is applied, the controller also checks planar base–box separation. Using support radii of the oriented box and torso along their center-to-center direction, it requires an additional 0.05 m clearance. A collision-free pose may be used as a **fallback-intermediate** plan even if its IK residual exceeds the CEM reachability threshold; such a plan is flagged for later replanning.

5.4.3 Asynchronous Reference Management

CEM runs on one background worker while MPPI continues using the current reference. A completed result is discarded if the request ID is obsolete, if the contact target moved more than 0.10 m, or if the box moved more than 0.10 m since submission. A refresh is requested when:

- engagement changes the contact target,
- the active target drifts by at least 0.08 m, or
- a fallback plan settles without becoming feasible.

Requests are separated by at least 10 controller steps, and fallback replanning is limited to eight attempts.

The accepted base reference is approached with rate limits. Height changes at at most 0.10 m/s. Horizontal speed is at most 1.0 m/s at standing height and decreases linearly to 40% of that value at the minimum height. Position and height tolerances use hysteresis to avoid repeatedly switching between walking and settled modes.

6 Nominal Whole-Body Trajectory

6.1 Height-Conditioned Leg Gait

The default and tracking gait is **stance_hold**; **walk_fast** is selected while planar base motion is required. Each gait has phase-compatible references at base heights

$$z_b \in \{0.35, 0.45, 0.543542\} \text{ m.} \quad (24)$$

The scheduler linearly interpolates joint positions and velocities between the two surrounding height levels. When the commanded height changes, the joint-velocity reference also includes the morphing derivative induced by $\dot{z}_b$. Gait switches preserve the previous phase instead of restarting the gait.

6.2 Arm IK Reference

At every controller update, the active moving contact target is converted to a four-joint arm reference with position-only damped least-squares IK:

$$\mathbf{e}^{(r)} = \mathbf{p}_e^* - \mathbf{p}_e(\mathbf{q}_a^{(r)}), \quad (25)$$

$$\Delta \mathbf{q}_a^{(r)} = \mathbf{J}_a^\top \left(\mathbf{J}_a \mathbf{J}_a^\top + \lambda_{\text{IK}}^2 \mathbf{I} \right)^{-1} \mathbf{e}^{(r)}, \quad (26)$$

$$\mathbf{q}_a^{(r+1)} = \text{clip} \left(\mathbf{q}_a^{(r)} + \alpha_{\text{IK}} \Delta \mathbf{q}_a^{(r)} \right). \quad (27)$$

The MPPI-side update uses damping 0.05, step size 0.7, and at most five iterations. The new solution is filtered with smoothing factor 0.3 and a per-update joint-change limit of 0.04 rad. The CEM reachability test uses the same IK structure but allows 40 iterations.

6.3 Whole-Body Prior and Residual Warm Start

The horizon reference is

$$\mathbf{r}_{j,k} = \begin{bmatrix} \mathbf{q}_{\ell,k}^g \\ \mathbf{q}_{\text{IK}} \\ \mathbf{q}_a \\ \mathbf{q}_{\ell,k}^g \\ \mathbf{0}_4 \end{bmatrix} \in \mathbb{R}^{32}. \quad (28)$$

Its position component plus a warm-started correction defines the nominal action:

$$\bar{\mathbf{u}}_k = \begin{bmatrix} \mathbf{q}_{\ell,k}^g \\ \mathbf{q}_{\text{IK}} \\ \mathbf{q}_a \end{bmatrix} + \Delta \mathbf{u}_k^{\text{warm}}. \quad (29)$$

After optimization, the selected correction relative to the gait/IK nominal is shifted by one step, and the new tail correction is set to zero. For the first 50 controller updates, the gait/IK reference is blended from the standing initialization.

7 Sampling and MPPI Update

7.1 Rollout Sampling Modes

The current controller exposes three main rollout modes: `original`, `original_spline`, and `safe_spline`. Both the controller constructor and `simulate_mppi.py` select `original_spline` by default. All three modes use $K = 4$ horizon-aligned knots

$$\mathcal{K} = \{\kappa_j\}_{j=0}^{K-1}, \quad \kappa_j = \text{round} \left(\frac{j(H-1)}{K-1} \right), \quad (30)$$

For $H = 40$, this gives $\mathcal{K} = \{0, 13, 26, 39\}$. The implementation requires the rounded indices to be unique. Let $\mathcal{S}_{\mathcal{K}}[\cdot]_k$ denote cubic-spline reconstruction at horizon index k from values defined at these knots. Each sample draws

$$\boldsymbol{\epsilon}_{n,j} \sim \mathcal{N}(\mathbf{0}, \text{diag}(\boldsymbol{\sigma}^2)), \quad j = 0, \dots, K-1. \quad (31)$$

For the gait-aware modes, define the full-rate gait/IK nominal $\mathbf{g}_k$, the shifted MPPI correction $\mathbf{c}_k$, and the clipped absolute warm start

$$\mathbf{a}_k = \text{clip}(\mathbf{g}_k + \mathbf{c}_k, \mathbf{u}_{\min}, \mathbf{u}_{\max}). \quad (32)$$

The three sampling rules are as follows.

original. This mode disables the explicit gait/IK nominal (`nominal_from_gait=False`) and maps internally to `sample_type=cubic_original`. Here $\mathbf{a}_k$ denotes the previous optimized absolute action sequence after receding-horizon shifting. The knot actions are perturbed and the complete absolute action is reconstructed:

$$\boxed{\mathbf{u}_{n,k}^{\text{original}} = \text{clip}(\mathcal{S}_{\mathcal{K}}[\mathbf{a}_{\kappa_j} + \boldsymbol{\epsilon}_{n,j}]_k, \mathbf{u}_{\min}, \mathbf{u}_{\max})}. \quad (33)$$

After the MPPI update, the selected absolute sequence is shifted and its last action is repeated at the new horizon tail.

original_spline. This is the default mode and maps internally to `sample_type=cubic_gait_absolute`. It enables the gait/IK nominal and forms $\mathbf{a}_k$ from the full gait/IK/residual warm start, but then splines that complete absolute trajectory:

$$\boxed{\mathbf{u}_{n,k}^{\text{original_spline}} = \text{clip}(\mathcal{S}_{\mathcal{K}}[\mathbf{a}_{\kappa_j} + \boldsymbol{\epsilon}_{n,j}]_k, \mathbf{u}_{\min}, \mathbf{u}_{\max})}. \quad (34)$$

Consequently, both the height-conditioned gait/IK component and the warm-started correction are reconstructed from only four knots. Short full-rate gait features between knots can therefore be smoothed.

safe_spline. This mode maps internally to `sample_type=cubic_gait_residual`. It preserves the complete full-rate gait/IK nominal and applies the cubic spline only to the correction plus exploration noise:

$$\mathbf{u}_{n,k}^{\text{safe_spline}} = \text{clip}(\mathbf{g}_k + \mathcal{S}_{\mathcal{K}}[\mathbf{c}_{\kappa_j} + \boldsymbol{\epsilon}_{n,j}]_k, \mathbf{u}_{\min}, \mathbf{u}_{\max}). \quad (35)$$

It is therefore the only one of the three modes that preserves the nominal gait/IK samples at every horizon step before final action clipping.

Candidate zero is overwritten with a mode-consistent noise-free baseline:

Mode	Candidate-zero baseline
original	$\mathbf{u}_{0,k} = \text{clip}(\mathbf{a}_k)$, the full shifted absolute warm start
original_spline	$\mathbf{u}_{0,k} = \text{clip}(\mathcal{S}_{\mathcal{K}}[\mathbf{a}_{\kappa_j}]_k)$
safe_spline	$\mathbf{u}_{0,k} = \text{clip}(\mathbf{g}_k + \mathcal{S}_{\mathcal{K}}[\mathbf{c}_{\kappa_j}]_k)$

Thus every batch contains a zero-noise reference appropriate to the selected sampling family. In **original_spline** and **safe_spline**, the selected correction relative to $\mathbf{g}_k$ is shifted after optimization and the new tail correction is set to zero.

The base exploration standard deviations are 0.03 for each hip, 0.10 for each thigh and calf, and 0.06rad for each arm joint. The **stance_hold** gait multiplies these values by 0.6; **walk_fast** uses scale 1.0.

7.2 Cost Normalization and Weights

Let $\mathcal{V}$ be the candidates that pass the hard arm-torso test and have finite total cost. For $n \in \mathcal{V}$,

$$S_{\min} = \min_{n \in \mathcal{V}} S_n, \quad (36)$$

$$\Delta S = \max_{n \in \mathcal{V}} S_n - S_{\min}, \quad (37)$$

$$w_n = \begin{cases} \exp\left[-\frac{1}{\lambda_{\text{MPPI}}} \frac{S_n - S_{\min}}{\Delta S}\right], & \Delta S > 10^{-12}, \\ 1, & \Delta S \leq 10^{-12}, \end{cases} \quad (38)$$

and $w_n = 0$ for invalid candidates. The configured temperature is $\lambda_{\text{MPPI}} = 0.1$. The updated sequence is the weighted average of the absolute candidate actions:

$$\mathbf{u}_k^+ = \text{clip}\left(\frac{\sum_{n=1}^N w_n \mathbf{u}_{n,k}}{\sum_{n=1}^N w_n + 10^{-10}}, \mathbf{u}_{\min}, \mathbf{u}_{\max}\right). \quad (39)$$

If every candidate is invalid, the last accepted safe trajectory is shifted and its final action is repeated. Only $\mathbf{u}_0^+$ is executed.

8 Cost Function

8.1 Robot State and Virtual PD Effort

The robot reference is assembled from the ramped base pose and velocity commands, the height-conditioned leg gait, and the arm IK reference. Before cost evaluation, the actual world-frame base linear velocity is transformed to the base-local frame.

Let $\mathbf{e}_{r,k}$ be the robot-state error after replacing the raw quaternion difference with base roll and pitch errors and zeroing the remaining two quaternion slots. The robot and virtual-effort cost

is

$$\tilde{\tau}_k = \mathbf{K}_p(\mathbf{u}_k - \mathbf{q}_{j,k}) - \mathbf{K}_d\dot{\mathbf{q}}_{j,k}, \quad (40)$$

$$\ell_{r,k} = \mathbf{e}_{r,k}^\top \mathbf{Q}_r \mathbf{e}_{r,k} + \tilde{\tau}_k^\top \mathbf{R} \tilde{\tau}_k. \quad (41)$$

The nonzero diagonal weights are:

- base $x/y/z$: 3000, 3000, and 1500,
- base roll/pitch: 5000 and 3000,
- all 16 joint positions: 500 each,
- base-local linear velocity x/y : 20 each,
- base linear z and all three angular velocities: 0.1 each,
- virtual PD effort: $\mathbf{R} = 10^{-3} \mathbf{I}_{16}$.

Yaw and all joint-velocity tracking weights are zero. The virtual effort does not include MuJoCo actuator-force saturation and therefore need not equal the force actually applied in the rollout.

8.2 Box Position and Tilt

The box position cost retains the weighted L1 form used by the push-box controller:

$$\boxed{\ell_{o,k} = 1000|p_{o,k}^x - p_o^{*,x}| + 1500|p_{o,k}^y - p_o^{*,y}|.} \quad (42)$$

The configured vertical box-position weight is zero, and box linear and angular velocities are not directly penalized.

Let $\mathbf{R}_{o,k}$ be the box rotation. Its yaw-invariant tilt angle is

$$\theta_{o,k} = \arccos(\text{clip}([\mathbf{R}_{o,k}]_{33}, -1, 1)). \quad (43)$$

The orientation cost is

$$\ell_{\text{tilt},k} = 10^4 \theta_{o,k}^2 + 10^5 \mathbb{I}\{\theta_{o,k} > 0.35\}. \quad (44)$$

The second term is a large soft discontinuous penalty; it does not mark a rollout invalid.

8.3 Moving EE Contact Cost

For the rollout-dependent target derived in Section 5,

$$\ell_{p,k} = 7000 \|\mathbf{p}_{e,k} - \mathbf{p}_{e,k}^*\|^2. \quad (45)$$

The implementation also supports the sign-invariant quaternion error

$$e_{R,k} = 2 \arccos(\text{clip}(|\mathbf{q}_{e,k}^\top \mathbf{q}_e^*|, -1, 1)), \quad \ell_{R,k} = w_R e_{R,k}^2, \quad (46)$$

but the push-box configuration sets $w_R = 0$. EE orientation tracking is therefore disabled.

The final EE term receives an additional multiplier of 10:

$$\Phi_{\text{EE}}(\mathbf{x}_H) = 10(\ell_{p,H-1} + \ell_{R,H-1}). \quad (47)$$

Because the same error is already present in the last stage, its total final coefficient is 11. There is no analogous terminal multiplier on the box position or tilt cost.

8.4 Arm–Torso Clearance

Four frames—shoulder, elbow, wrist, and EE—form three arm capsules with radii

$$[0.030, 0.030, 0.0375] \text{ m.} \quad (48)$$

The torso is the oriented box associated with `base_collision`, with half-size $[0.25, 0.14, 0.075]$ m. The proximal 0.07 m of the first capsule is excluded near the structural shoulder mount.

For signed capsule clearances $d_{k,s}$, the soft cost is

$$\ell_{c,k} = 10^4 \sum_{s=1}^3 [\max(0.005 - d_{k,s}, 0)]^2. \quad (49)$$

A complete rollout is valid only if

$$d_{k,s} \geq 0 \quad \text{for every } k \text{ and } s. \quad (50)$$

Thus capsule penetration is rejected, while a tangent configuration is allowed. The shared collision helper first uses uniformly sampled centerline points to bound clearance and invokes the exact segment-to-box distance only when needed for hard-validity classification.

8.5 Total Stage and Rollout Cost

The complete stage cost is

$$\ell_k = \ell_{r,k} + \ell_{o,k} + \ell_{\text{tilt},k} + \ell_{p,k} + \ell_{R,k} + \ell_{c,k}. \quad (51)$$

Candidate n has total cost

$$S_n = \sum_{k=0}^{H-1} \ell_k^{(n)} + 10 \left(\ell_{p,H-1}^{(n)} + \ell_{R,H-1}^{(n)} \right). \quad (52)$$

Only arm–torso hard clearance and non-finite cost affect the validity mask; large box tilt remains part of S_n rather than causing hard rejection.

9 Complete Receding-Horizon Algorithm

The standard `simulate_mppi.py` configuration uses a 0.01 s simulation step and a 100 Hz controller rate, so one MPPI update and one task-success check are requested per simulation step.

Algorithm 1 `push_box` CEM-guided whole-body MPPI update

Require: Observation $\hat{\mathbf{x}}$, active base plan, previous safe MPPI plan

- 1: Split $\hat{\mathbf{x}}$ into robot and box states.
 - 2: Compute the box-to-goal direction and moving EE contact target.
 - 3: Engage the penetration target if the EE reaches the pre-contact target.
 - 4: Apply a completed non-stale CEM plan, if available.
 - 5: Submit a background CEM refresh when required and allowed by cooldown.
 - 6: Ramp the base reference toward the accepted CEM pose.
 - 7: Select `walk_fast` during planar motion; otherwise select `stance_hold`.
 - 8: Solve smoothed position IK for the current moving EE target.
 - 9: Combine height-conditioned gait and arm IK with the shifted MPPI correction.
 - 10: Generate $N = 30$ candidates using the selected `original`, `original_spline`, or `safe_spline` rule.
 - 11: Preserve one mode-consistent noise-free baseline candidate.
 - 12: Roll out every candidate for $H = 40$ MuJoCo steps with FK sensors.
 - 13: Evaluate robot, box, EE-contact, tilt, and arm-clearance costs.
 - 14: Set zero weight for every hard-collision or non-finite rollout.
 - 15: **if** at least one valid candidate exists **then**
 - 16: Normalize valid costs and compute exponential weights.
 - 17: Take the weighted mean of absolute candidate actions and save it as the latest safe plan.
 - 18: **else**
 - 19: Shift and reuse the previous safe plan.
 - 20: **end if**
 - 21: Shift either the absolute plan (`original`) or the optimized gait/IK correction (gait-aware modes), and advance gait phase.
 - 22: **return** The first desired joint-position command $\mathbf{u}_0^+$
-

10 Default Configuration Summary

Parameter	Default value
Robot / object	B2 with four-DoF Z1 / free 0.38 m box
Full state / input dimension	58 / 16
Robot / box state dimension	45 / 13
Input interpretation	desired joint position
Time step / horizon	0.01 s / 40 steps (0.4 s)
MPPI samples / workers	30 / 5
MPPI temperature	0.1
Rollout mode / knots	<code>original_spline</code> (default) / four
Initial keyframe	<code>stand</code>
Tracking / approach gait	<code>stance_hold</code> / <code>walk_fast</code>
Height-conditioned levels	0.35, 0.45, 0.543542 m
CEM variable	$[\Delta x_b, \Delta y_b, z_b]^\top$
CEM samples / iterations / elite fraction	30 / 3 / 0.20
CEM maximum horizontal shift	0.40 m
Box position target	$[3.0, 2.0, 0.19]^\top$ m
Pre-contact gap / push penetration	0.05/0.015 m
EE position/orientation weights	7000 / 0
EE terminal scale	10
Box $x/y/z$ L1 weights	1000 / 1500 / 0
Box tilt weight / threshold / violation penalty	10^4 / 0.35 rad / 10^5
Collision safe/hard distances	0.005/0.0 m
Collision soft weight	10^4
Box success tolerance / hold	0.20 m / 20 checks

12 Implementation Caveats and Limitations

1. **The update differs from canonical MPPI.** Costs are normalized by the valid batch’s min–max range, and absolute candidate actions are averaged. There is no control-noise correction term. Consequently, $\lambda = 0.1$ controls selection pressure over normalized relative cost rather than temperature in the original physical cost units.
2. **Engagement is geometric, not contact-sensor based.** The target switches after an EE position-tolerance test. It does not verify normal force, sustained contact, or whether the EE is on the intended face.
3. **The CEM planner is kinematic and local.** It searches at most 0.40 m from the base pose at request time, assumes an upright base, and does not predict balance, gait feasibility, box motion, or dynamic contact. A fallback pose may intentionally remain outside the configured IK residual threshold.
4. **The moving contact target is a point objective.** EE orientation has zero weight. The point is placed at a fixed vertical offset from the box center and does not optimize contact normal alignment, friction cone margin, or wrench direction.
5. **Box tilt is not a hard constraint.** Exceeding 0.35 rad adds 10^5 to the cost but does not invalidate the rollout. Whether this is sufficient depends on the scale and spread of all other accumulated costs.

6. **The collision model is incomplete.** Hard rejection covers three arm capsules against one torso box. It does not guarantee arm–leg, gripper–environment, box–base, or general self-collision safety over MPPI rollouts. The separate CEM acceptance check covers only the planned base footprint against the current box footprint.
7. **Analytic and physical collision models are intentionally different.** Arm–torso physical contact is disabled in the rollout model while the analytic capsule constraint is active. Model mismatch in capsule radii, torso geometry, or exclusions directly affects safety.
8. **Effort cost is not applied torque.** The virtual PD effort omits actuator-force saturation. Large errors can therefore make its numerical penalty differ from MuJoCo’s applied force.
9. **Several quantities are soft-only.** The box target, box uprightness, base tracking, IK reachability during MPPI, and EE contact are objectives or references, not formal dynamic constraints.
10. **Real-time performance must be measured.** Each update evaluates 30×40 contact-dynamics steps and rollout sensors, while CEM occasionally evaluates 90 IK/collision candidates on a background thread. Sustained 100 Hz operation depends on the target CPU and thread contention.

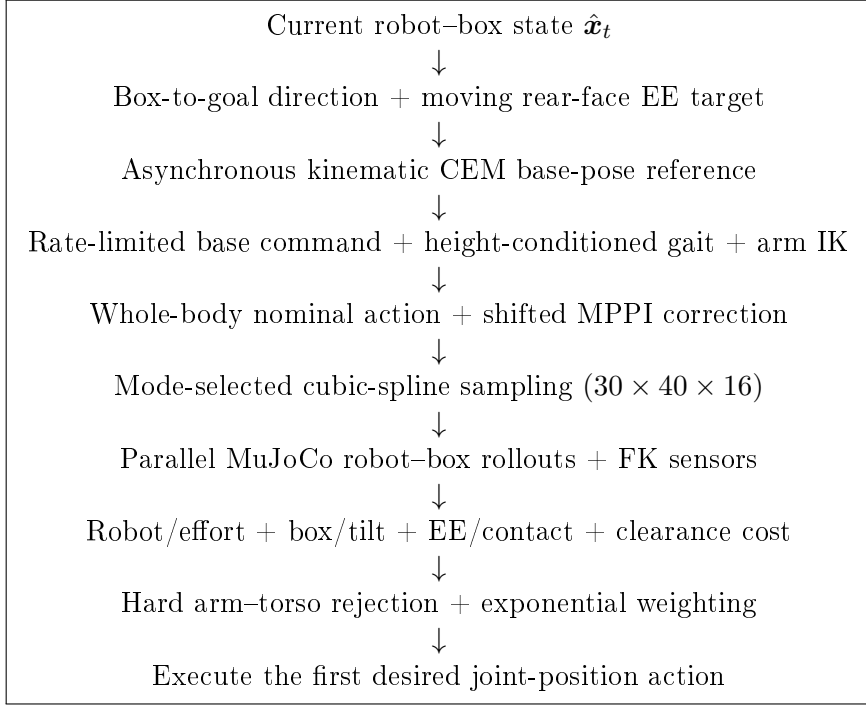
13 Recommended Validation

Useful closed-loop measurements include:

1. mean, 95th, and 99th percentile MPPI update time and the 10 ms deadline-miss rate, with and without concurrent CEM,
2. CEM latency, IK residual, accepted/fallback/rejected plan counts, and stale-plan discard count,
3. box x/y error, path length, speed, yaw, tilt, and completion time,
4. EE-to-moving-target error, engagement time, and measured EE–box contact force,
5. minimum arm–torso clearance, valid-rollout fraction, and all-invalid MPPI fallback count,
6. base tracking error, roll/pitch, foot contacts, and slip,
7. gait/IK nominal, selected actions, warm-start correction, and actuator saturation rate,
8. one-step error between predicted and executed robot and box states.

Useful ablations include disabling the CEM base planner, removing the penetration switch, replacing the geometric engagement test with contact feedback, removing the tilt violation penalty, disabling terminal EE emphasis, and comparing `stance_hold` noise scale 0.6 against 1.0.

14 Pipeline Summary



The defining design choice is the separation between a low-dimensional kinematic CEM reference planner and a full-dynamics MPPI controller. CEM moves the base reference toward a posture from which the evolving contact point is reachable; MPPI then searches whole-body desired joint-position trajectories whose MuJoCo rollouts actually stabilize the robot and move the free box.